

Set-up and registration

International Marketing Analytics



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[Office hours: book online \(Google Calendar\)](#)

Today: from zero to a working set-up

Accounts

Tools

Repo

Packages

Check

AI

What you need



- Your laptop and charger
- Windows 10/11 or a recent macOS
- About 10 GB free disk space
- Admin rights to install software

Have at hand



- Your WU e-mail and its password
- Your phone, for two-factor sign-in
- Your student ID or enrolment confirmation

How we work



- One step at a time, all together
- Check your neighbour's screen
- Stuck? Raise your hand, keep going with the next step



Goal: by the end of today the check script prints “ready” on every laptop.

Register your accounts



Part 1 of 4 · GitHub, GitHub Education,
DataCamp

Three registrations

Accounts

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GitHub

Stores code and its history;
hosts the course and group
repositories

github.com/signup



GitHub Education

Free student benefits,
including GitHub Copilot

education.github.com/pack



DataCamp

Online Python courses; four
certificates count 20 %

Invitation link on Canvas



Same e-mail everywhere: your WU student address unlocks the free plans.

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1 Open github.com/signup

2 **WU e-mail:** sign up with your student address

3 **Username:** short and professional, e.g. anna-berger

4 **Verify** the code GitHub sends to your inbox

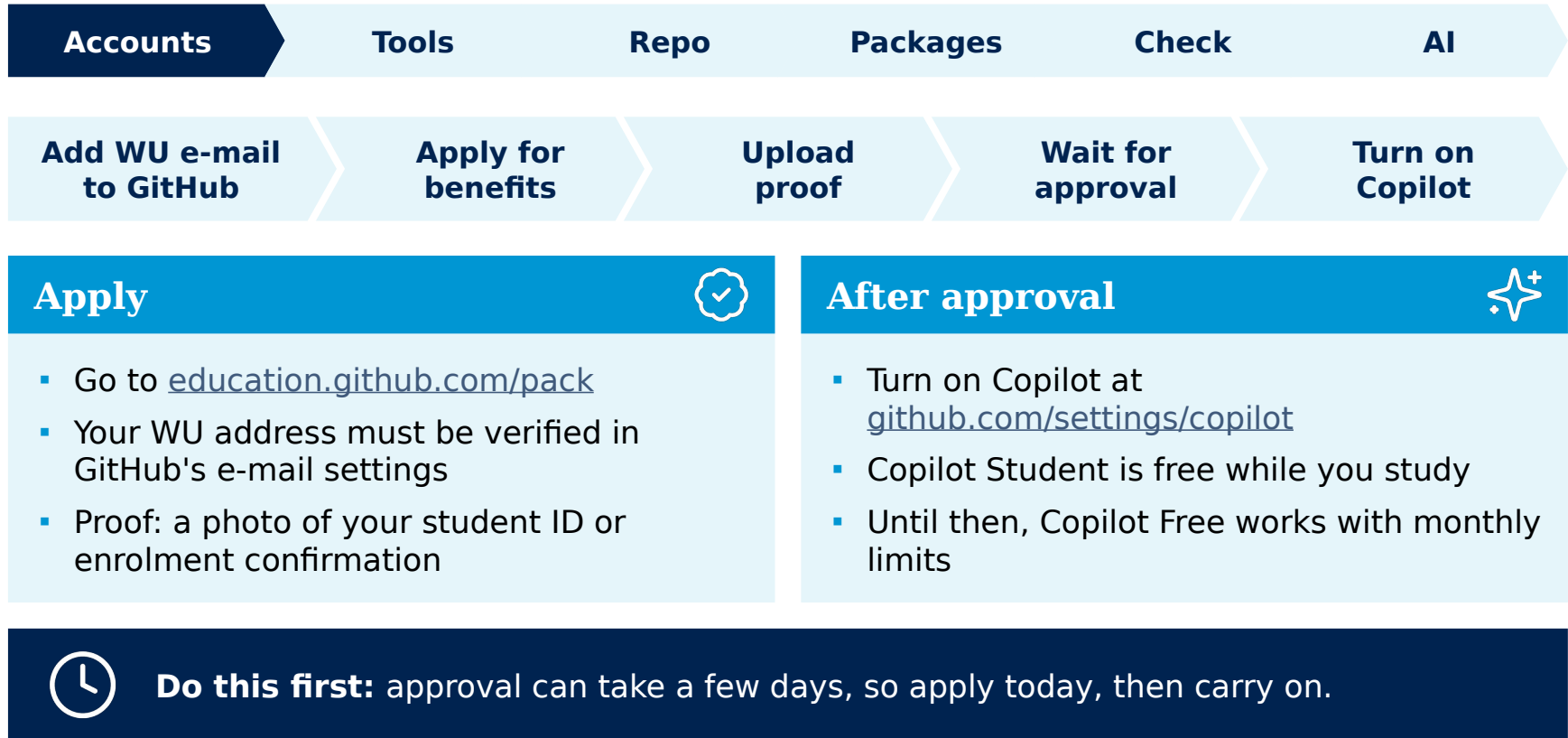
5 **Two-factor sign-in:** GitHub Mobile or an authenticator app

Why it matters



- Your username appears on every commit you make
- A clean GitHub profile is part of your CV
- You need it for the group repository of the case study
- The free plan is all you need

GitHub Education and Copilot



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1 **Open** the invitation link on Canvas

2 **Sign up** with your WU e-mail, or log in

3 **Join** the classroom International Marketing Analytics

4 **Find** the four assigned courses under Assignments

5 **Upload** each certificate to Canvas by its deadline

6

months free premium access

4

courses with certificates

20 %

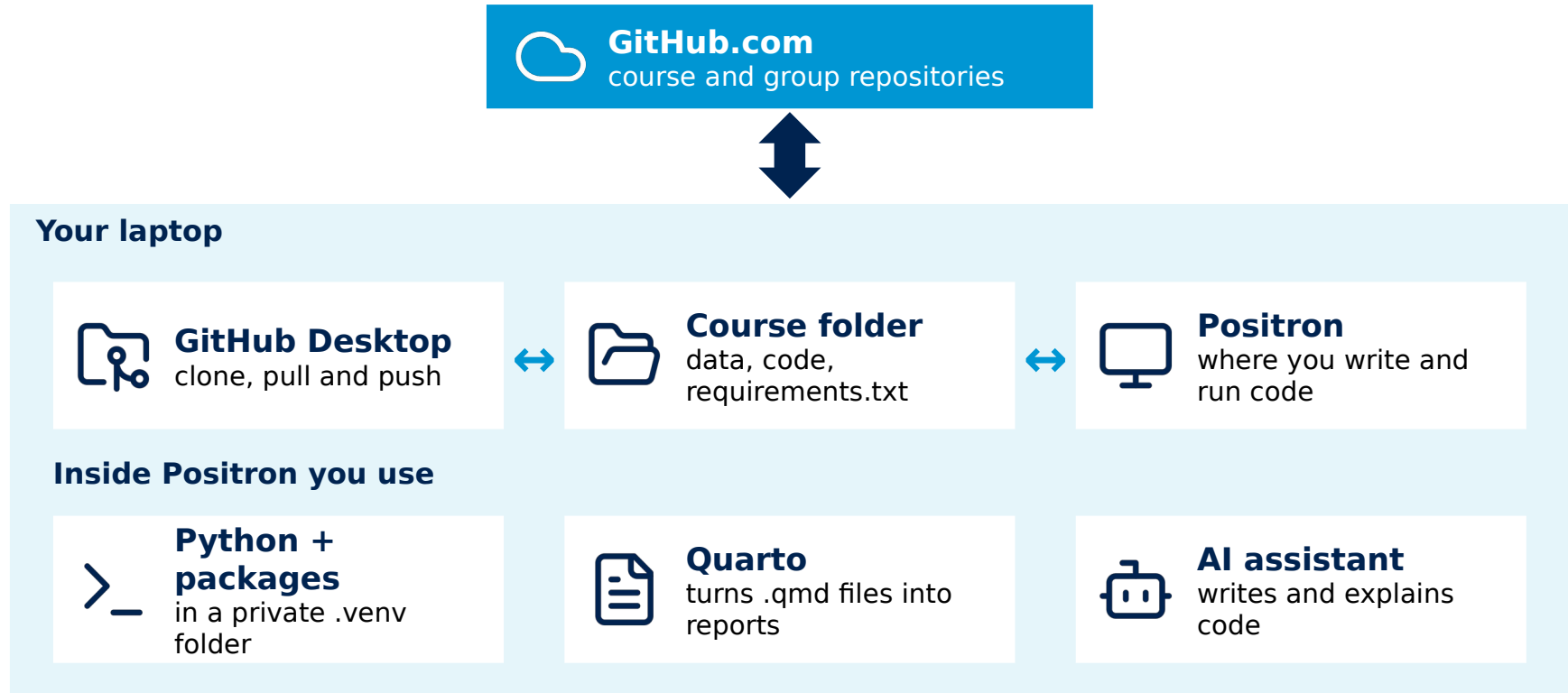
of the final grade

Install the tools



Part 2 of 4 · Python, Positron, Quarto, GitHub
Desktop, the course repository

How the pieces fit together



Step 2: download and install Python 3.13.16

Accounts

Tools

Repo

Packages

Check

AI

Windows



- **Download:** Windows installer (64-bit)
- Run the downloaded .exe
- Tick “Add python.exe to PATH”
- Click Install Now

```
py -3.13 --version
```

macOS



- **Download:** macOS 64-bit universal2 installer
- Open the downloaded .pkg
- Click through the installer
- Works on Apple silicon and Intel

```
python3.13 --version
```



Both installers: python.org/downloads/release/python-31316

Not the yellow “Download Python 3.14” button: the course needs 3.13.

Step 2: install Positron, Quarto and GitHub Desktop

Accounts

Tools

Repo

Packages

Check

AI

Positron



- The editor for code, data and charts
- Version 2026.09, free
- Welcome page checks that Python is ready

positron.posit.co/download

Quarto



- Turns text and code into reports and slides
- Check in a terminal:
`quarto --version`

quarto.org/docs/get-started

GitHub Desktop



- Clone, pull and push with clicks
- Installs Git for you
- Sign in with your GitHub account

desktop.github.com



Restart Positron after installing Quarto, so it finds it.

Step 3: get the course repository

Accounts

Tools

Repo

Packages

Check

AI

Copy the URL
from Canvas

GitHub Desktop:
Clone repository

Pick a folder
without spaces

Positron:
Open Folder

Good folder names

C:\Users\anna\wu\ima
~/wu/ima

Avoid spaces, umlauts and cloud folders such as OneDrive or iCloud: they break paths and syncing.

Clone: your own copy of the repository, linked to GitHub

Pull: get the latest materials; do it before every session

Commit and push: save and upload your changes (group repository)



In Positron: File → Open Folder and choose the cloned folder.

Step 4: install the course packages

Accounts

Tools

Repo

Packages

Check

AI

Windows: Positron terminal (PowerShell)

```
py -3.13 -m venv .venv  
.venv\Scripts\python -m pip install -r requirements.txt
```

macOS: Positron terminal

```
python3.13 -m venv .venv  
.venv/bin/python -m pip install -r requirements.txt
```

Line 1 creates .venv, a private Python for this folder.

Line 2 installs all course packages into it.

Then: Interpreter picker

Python 3.13.16 (.venv) ▼

- Top right in Positron
- Pick the Python in .venv
- Positron remembers it for this folder



No typing? Command Palette → Python: Create Environment → Venv → tick requirements.txt.

Step 5: run the check

Accounts

Tools

Repo

Packages

Check

AI

Windows

```
.venv\Scripts\python setup\check_setup.py
```

macOS

```
.venv/bin/python setup/check_setup.py
```

Example output (shortened)

```
Packages
[ok] polars          1.44.2    data wrangling
[ok] plotnine        0.15.8    charts
[ok] great_tables    1.0.0     tables
[ok] statsmodels      0.15.0    regression
...
ready
```

Reading it



- [ok]: in place
- [!]: the line says what to fix
- Ends with “ready”: done



No terminal? Open setup/check_setup.py in Positron and press Run.

Step 6: connect your AI assistant

Accounts

Tools

Repo

Packages

Check

AI

- 1 Command Palette:** Configure Language Model Providers
- 2 Choose GitHub Copilot** and sign in with GitHub
- 3 Open the chat:** View: Show Posit Assistant

Providers

GitHub Copilot: chat and completions; free with Copilot Student or Copilot Free

Posit AI Pass: chat and completions; about USD 20 a month

Anthropic API key: chat; pay as you go



Our rule: the AI writes, you check. Run the code, read the output and keep a short prompt log for each submission.

How Positron works



Part 3 of 4 · panes, running code, projects,
Quarto, Posit Assistant

New ▼ Open ▼

case-study ▼

Python 3.13.16 (.venv) ▼

1 Explorer
files of the open folder

data/
analysis.py
report.qmd
requirements.txt

2 Editor
write .py and .qmd files

%%
sales = pl.read_csv(...)
sales.head()

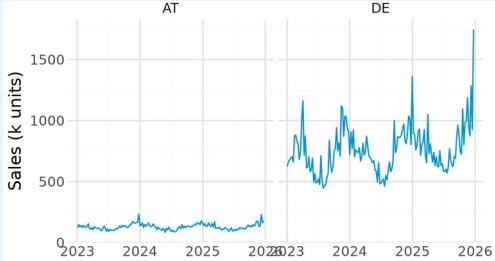
3 Console and Terminal
the Panel: code runs, results print

>>> sales.height
936

4 Variables
Session pane: what is in memory

sales 936 × 18
summary 6 × 3
model OLS results

5 Plots
Session pane: your charts



From code to result

**Write code
in the editor**

**Ctrl/Cmd
+ Enter**

**It runs in
the console**

**Objects in
Variables**

**Charts in
Plots**

Run code



- Ctrl/Cmd + Enter: the selection or current statement
- Ctrl/Cmd + Shift + Enter: the whole file, or the current cell
- # %% starts a cell in a .py file

Look at results



- Variables: every object, its type and size
- Data Explorer: click a table, or type %view sales
- Plots: earlier charts, zoom, save as PNG



Console or terminal? The console speaks Python. The terminal speaks to your computer: pip, git and quarto commands go there.

Projects in Positron: one folder for everything

A project is a folder

```
case-study/
├── .venv/           private Python + packages
├── .git/            version history
├── .gitignore       what Git leaves out
├── data/            raw data, never edit by hand
├── analysis.py      code, split into # %% cells
├── report.qmd       text, code and results
└── requirements.txt packages and versions
```

Open the folder, not single files: File
→ Open Folder

Positron remembers the interpreter
and open files per folder

Relative paths: data/sales.csv works
on every laptop

Switch projects with the project
switcher, top right



.venv stays on your laptop: .gitignore keeps it out of GitHub. requirements.txt travels instead, so your group rebuilds the same environment with one command.

Start a new project from a template

New Folder
from Template

Python
Project

Name and
location

Python 3.13
+ venv

Create

Positron sets up



- The folder with a .venv inside
- Git: .git and a .gitignore
- A running Python session
- An empty file to start with

Then, in the Positron console

```
%pip install polars plotnine great-tables
```

```
%pip freeze > requirements.txt
```

Line 1 installs packages, line 2 writes their versions to requirements.txt.



Where: New ▼ (top left), the project switcher, or Command Palette → Workspaces: New Folder from Template.

Your case study as a shared project

**github.com:
new repository**

**Invite
the group**

**Desktop:
clone**

**Positron:
Open Folder**

**Create
.venv**

github.com



- One member creates the repository:
Python .gitignore,
README
- Settings → Collaborators:
invite the group

GitHub Desktop



- Pull before you start
- Commit small steps with
a clear message
- Push when you stop

Positron



- Open the cloned folder
- Create .venv from
requirements.txt (Step 4)
- Work, save, then back to
Desktop



Fewer conflicts: pull first, push often, and one person per file where you can.

Quarto documents in Positron

report.qmd

```
---  
title: "Sales by country"  
format: html  
---  
  
## Revenue in 2025  
  
``{python}  
import polars as pl  
sales = pl.read_csv("data/mmm/alpenglow_weekly.csv")  
``  
  
Germany has the highest revenue.
```



Text in Markdown: headings, bold, lists



Code cells run one by one; output shows below



Preview: Ctrl/Cmd + Shift + K shows the report next to your code



Render to HTML, PDF, Word or slides



One file, rerun any time: the labs and your case study report are Quarto documents.

Shortcuts worth knowing

Action	Windows	macOS
Run selection or current statement	Ctrl + Enter	Cmd + Enter
Run the file or current cell	Ctrl + Shift + Enter	Cmd + Shift + Enter
Insert a code cell (# %%)	Ctrl + Shift + I	Cmd + Shift + I
Restart the Python session	Ctrl + Shift + 0	Cmd + Shift + 0
Clear the console	Ctrl + L	Ctrl + L
Preview a Quarto document	Ctrl + Shift + K	Cmd + Shift + K
Command Palette	Ctrl + Shift + P	Cmd + Shift + P
Help for the word under the cursor	F1	F1
Accept an AI suggestion	Tab	Tab

Posit Assistant: the AI inside Positron

What it sees 	What it does 	How you steer it 
▪ Your open files and project ▪ Loaded data, variables and plots ▪ Console history	▪ Explains, writes and fixes code ▪ Runs code after you approve ▪ Fix and Explain on any error	▪ <code>/plan</code>: a plan before any code ▪ <code>@file</code>: point it to a file ▪ Permission mode: keep Normal

Open it

View: Show Posit Assistant

Command Palette, or the Assistant icon in the Activity Bar

Memory: an AGENTS.md file in your project tells it your conventions, for example “use polars and plotnine”.

Completions: ghost text



- Grey suggestions appear as you type
- Tab accepts, Ctrl/Cmd + → word by word
- Esc or keep typing to dismiss

Chat: Posit Assistant



- Ask in plain English, one step at a time
- Check every result it reports
- Export the chat: your prompt log

Good prompts

"Sum revenue by country for 2025 with polars and sort it."
"Explain this error in two sentences, then fix it."



Privacy: prompts, variable names and types go to the model provider; rows of data only if you ask. Never paste personal data.

Python packages for this course



Part 4 of 4 · install and load; polars, plotnine,
Great Tables, statsmodels

Install packages: three ways

1 Windows: Positron terminal

```
.venv\Scripts\python -m pip install -r requirements.txt # all course packages  
.venv\Scripts\python -m pip install polars             # one package
```

2 macOS: Positron terminal

```
.venv/bin/python -m pip install -r requirements.txt    # all course packages  
.venv/bin/python -m pip install polars                 # one package
```

3 Windows and macOS: Positron console

```
%pip install polars plotnine great-tables
```



Install name and import name can differ: pip install great-tables → import great_tables; scikit-learn → sklearn; pymc-marketing → pymc_marketing.

The Packages pane: install without code

**Packages icon
(Activity Bar)**

**Install
Package**

**Search name,
pick version**

**Restart session
when asked**

PACKAGES Python 3.13.16 (.venv)

great-tables	1.0.0	
pandas	3.0.6	● attached
plotnine	0.15.8	
polars	1.44.2	● attached
statsmodels	0.15.0	● attached
...		

What else it does



- Shows which packages are loaded (attached)
- Update one, or Update All Packages
- Flags known security issues
- Keeps versions in line with requirements.txt
- Offers to install a missing package after an error

Load packages: import at the top of every file

**Install once
per laptop**

**Import in every
file and session**

**Use the short
name: `pl.read_csv()`**

Windows and macOS: the same code

```
import polars as pl
import statsmodels.formula.api as smf
from plotnine import ggplot, aes, geom_line
from great_tables import GT

print(pl.__version__)    # 1.44.2
```

Conventions

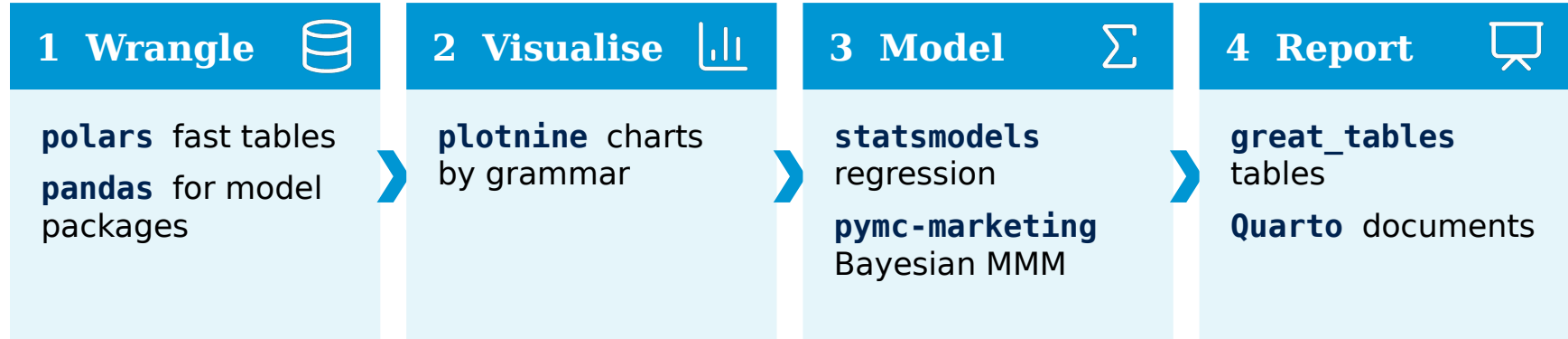


- `pl` and `smf` are the usual short names
- Imports go at the top
- Import only what you use



ModuleNotFoundError? Either the wrong interpreter is selected (pick the one in `.venv`) or the package is missing: `%pip install it`.

Your toolkit, from data to report



22

packages pinned in requirements.txt

1

command installs them all

0

euros: all open source



Online you will also meet pandas and matplotlib: older but common; the ideas carry over.

Revenue and TV spend per country, 2025

```
import polars as pl

sales = pl.read_csv("data/mmm/alpenglow_weekly.csv",
                    try_parse_dates=True)

summary = (
    sales
    .filter(pl.col("week").dt.year() == 2025)
    .group_by("country")
    .agg(pl.col("revenue_eur_k").sum().alias("revenue"),
         pl.col("spend_tv_k").sum().alias("tv"))
    .sort("revenue", descending=True)
)
```

DataFrame: a table with typed columns

Expressions: `pl.col("tv").sum()`
says what to compute

Chaining: one step per line, read top to bottom

Fast: multi-threaded, written in Rust

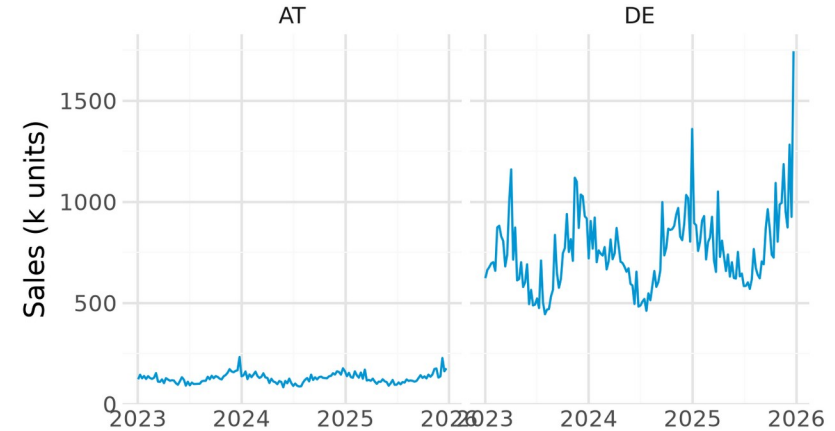


Result: six rows, one per country. Germany leads with about EUR 131 million revenue in 2025. Need pandas? Add `.to_pandas()`.

plotnine: charts with the grammar of graphics

Weekly sales, Austria and Germany

```
(ggplot(sales.filter(
  pl.col("country").is_in(["AT", "DE"])),
  aes(x="week", y="sales_units_k"))
+ geom_line(color="#0096D3")
+ facet_wrap("country")
+ scale_x_date(date_breaks="1 year",
  date_labels="%Y")
+ labs(x="", y="Sales (k units)")
+ theme_minimal())
```



The output of the code on the left

ggplot()
the data

aes()
columns to x, y

geom_
the marks

facet_
one panel per
group

theme_
the look

Data + mapping + marks + panels + theme: each + adds one layer.

Great Tables: tables for reports

The polars summary as a table

```
from great_tables import GT

(GT(summary)
 .tab_header(title="Revenue and TV spend, 2025")
 .fmt_number(columns=["revenue", "tv"], decimals=0)
 .cols_label(country="Country",
              revenue="Revenue (k EUR)",
              tv="TV spend (k EUR)"))
```

Header and labels: readable names instead of column codes

Number formats: thousands separators, currency, percent

Works with polars and pandas; renders in Quarto reports

Revenue and TV spend, 2025		
Country	Revenue (k EUR)	TV spend (k EUR)
DE	130,650	5,434
FR	66,306	2,363
IT	47,950	2,478
NL	33,234	812
PL	23,112	616
AT	21,984	695

Weekly sales in Austria, 2023 to 2025

```
import statsmodels.formula.api as smf

at = sales.filter(pl.col("country") == "AT").to_pandas()
model = smf.ols(
    "sales_units_k ~ price_eur + spend_tv_k"
    " + spend_paid_search_k",
    data=at).fit()
print(model.summary())
```

Coefficient	Estimate
Intercept	250.63
price_eur	-48.31
spend_tv_k	0.41
spend_paid_search_k	4.03
R ² (n = 156)	0.43



Reading it: EUR 1,000 more paid search in a week goes with about 4,000 more units sold, holding price and TV constant.

Later in the course: scikit-learn for logistic regression and pymc-marketing for Bayesian MMM.

When something goes wrong

Problem	Fix
Windows: python or py is not recognised	Reinstall Python 3.13 with “Add python.exe to PATH” ticked
No .venv in the Interpreter picker	Open the course folder, not a single file; then pick Python 3.13.16 (.venv)
ModuleNotFoundError	Select the .venv interpreter, then %pip install the package
macOS: externally-managed-environment	You used the system Python: use the .venv/bin/python commands
Quarto preview does not start	Restart Positron; run quarto check in the terminal
Clone fails in GitHub Desktop	Sign in again under Accounts in the settings; check the URL
Copilot is not offered	GitHub Education still pending: use Copilot Free meanwhile

Checklist: before you leave today

Accounts



- ☐ GitHub account with two-factor sign-in
- ☐ GitHub Education application sent
- ☐ DataCamp classroom joined

Laptop



- ☐ Python 3.13, Positron, Quarto, GitHub Desktop
- ☐ Course repository cloned and open
- ☐ .venv created and selected
- ☐ Check script prints “ready”
- ☐ Posit Assistant connected



Not ready yet? Post the failing step in the Q&A forum on Canvas before Session 2.



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